

BI Strategy Document

Global Logistics Business Intelligence & Analytics

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Project: Global Logistics Business Intelligence & Analytics

Technology Stack: Python, Pandas, PostgreSQL, SQL, Tableau

1. Strategy Overview

The Global Logistics BI strategy establishes a structured approach for transforming logistics data into reliable and actionable business insights.

The solution combines **Python and Pandas** for data preparation and transformation, **PostgreSQL** for centralized analytical storage, **SQL** for analysis and validation, and **Tableau** for interactive Business Intelligence dashboards. The strategy is designed to connect business requirements with data architecture, KPI development, analytical processes, and dashboard reporting.

2. Business & BI Objectives

The primary objectives of the BI strategy are to:

- Create a centralized view of logistics operations.
 - Improve visibility into shipment performance.
 - Monitor route, carrier, driver, and vehicle performance.
 - Analyze customer and financial performance.
 - Identify operational bottlenecks and shipment delays.
 - Understand the impact of weather and road conditions.
 - Provide decision-ready information for management and operational teams.
 - Establish a scalable foundation for future predictive and real-time analytics.
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3. Data Strategy

The data strategy focuses on creating a reliable and consistent analytical foundation.

The project uses Python and Pandas to prepare and transform data before loading it into the PostgreSQL analytical environment.

Data Preparation

The data preparation process includes:

- Data cleaning.
- Data transformation.
- Data type validation.
- Handling missing or inconsistent values.
- Standardization of business attributes.
- Preparation of analytical measures.
- Validation of transformed data.

Data Integration

The solution integrates multiple business areas, including:

- Shipments
- Customers
- Products
- Routes
- Carriers
- Drivers
- Vehicles
- Warehouses

- Weather
- Dates

This integrated structure enables stakeholders to analyze logistics performance from multiple business perspectives.

4. Data Warehouse Strategy

The project uses an **Enterprise Star Schema** to organize the analytical data.

Fact tables contain measurable business events and performance metrics, while dimension tables provide descriptive business context.

The primary dimensions include:

- dim_date
- dim_customers
- dim_products
- dim_routes
- dim_carriers
- dim_drivers
- dim_vehicles
- dim_warehouses
- dim_weather

The fact tables provide the central measurements required for shipment and logistics analysis.

Key Design Principles

- Use clear primary and foreign key relationships.
 - Separate descriptive attributes from measurable facts.
 - Maintain reusable dimension structures.
 - Support scalable analytical queries.
 - Maintain consistent business definitions.
 - Enable efficient Tableau reporting.
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5. Analytics & KPI Strategy

KPIs are selected based on stakeholder requirements and the business questions identified during the project.

Each KPI should have:

- A clear business definition.
- A consistent calculation method.
- Appropriate aggregation.
- Consistent formatting.
- A clear connection to a business decision.

Key KPIs

KPI	Purpose
Total Shipments	Monitor overall shipment activity.
Total Routes	Understand network size and route coverage.
Avg Transit Time	Evaluate transportation efficiency.
Avg Delay Time	Monitor delivery performance.
Top Route by Shipments	Identify high-volume routes.
Avg Delay by Route	Identify routes with potential bottlenecks.
Driver Performance	Evaluate driver capability and performance.
Vehicle Efficiency & Capacity	Assess fleet utilization and efficiency.
Weather Impact	Understand environmental effects on shipment delays.

6. Dashboard & Visualization Strategy

The Tableau dashboards are designed around **business questions rather than individual data fields**.

The visualization strategy uses different chart types depending on the analytical purpose.

Visualization Purpose

- KPI Cards** Fast executive-level performance monitoring.
- Bar Charts** Compare categories, routes, and performance rankings.
- Line Charts** Analyze trends over time.
- Maps** Analyze geographic and route-related information.
- Scatter Plots** Analyze relationships between two numerical variables.
- Donut Charts** Show simple part-to-whole distributions.

Dashboard Design Principles

- Use a consistent **1200 × 900 fixed dashboard size**.
 - Keep KPI cards clearly visible.
 - Use clear and business-friendly titles.
 - Maintain consistent formatting across dashboards.
 - Use blue as the primary analytical color.
 - Use contrasting colors for delays, risks, or critical conditions.
 - Avoid unnecessary visual clutter.
 - Use filters to support interactive analysis.
 - Use tooltips to provide additional context.
 - Keep charts readable and appropriately spaced.
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7. Dashboard Strategy

The project includes seven dashboards, each addressing a specific business area.

Dashboard 1 — Executive Overview

Provides a high-level summary of overall logistics performance using key KPIs and trends.

Dashboard 2 — Operations Performance

Focuses on shipment activity, delivery performance, transit time, delays, and operational trends.

Dashboard 3 — Financial & Cost Analysis

Provides visibility into financial and logistics cost performance.

Dashboard 4 — Customer Revenue & Segmentation

Analyzes customer contribution, revenue distribution, and customer segments.

Dashboard 5 — Driver & Vehicle Performance

Evaluates driver performance, driver experience, vehicle efficiency, fuel consumption, and vehicle capacity.

Dashboard 6 — Weather & Shipment Impact

Analyzes the relationship between weather conditions, weather severity, road conditions, and shipment delays.

Dashboard 7 — Route Performance & Network

Analyzes route volume, average delays, geographic distribution, and the relationship between shipment volume and route performance.

8. Data Quality & QA Strategy

Data quality is an important part of the BI strategy because dashboard decisions depend on the reliability of the underlying data.

The project applies several validation steps:

- Validate database schema and table structures.
- Validate primary and foreign key relationships.
- Check data types.
- Review missing and inconsistent values.

- Validate calculated fields.
- Verify KPI definitions.
- Compare dashboard results with underlying data.
- Test filters and dashboard interactions.
- Review chart labels and units.
- Check dashboard readability.
- Review tooltips.
- Perform final dashboard Quality Assurance.

The final Tableau dashboards were reviewed before completion to ensure that the visualizations and KPIs behaved as expected.

9. Deployment & Portfolio Strategy

The project will be organized as a portfolio-ready BI solution.

The recommended project structure is:

Global-Logistics-BI/

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|—— README.md
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|—— Documentation/
|   |—— Business_Requirements/
|   |—— Data_Dictionary.xlsx
|   |—— Project_Documentation.pdf
|   |—— Data_Model.png
|
|
|—— Data/
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|
|—— SQL/
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|
|—— Python/
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|
|—— Tableau/
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|
|—— Screenshots/
```

Deployment Principles

- Maintain project code and documentation in GitHub.
- Publish Tableau dashboards through Tableau Public where appropriate.
- Avoid publishing sensitive or confidential data.
- Avoid uploading unnecessarily large raw datasets.
- Include dashboard screenshots for portfolio presentation.
- Maintain a clear README explaining the project and its business value.

10. Maintenance Strategy

The BI solution should be maintained as business requirements and source data evolve.

Maintenance activities include:

- Updating data preparation scripts when source structures change.
- Reviewing KPI definitions when business requirements change.
- Monitoring data quality.
- Monitoring dashboard performance.
- Updating Tableau dashboards when new business requirements emerge.
- Documenting major changes to the data model.
- Maintaining project versions through Git.

11. Future Improvements

The current solution provides a strong foundation for future BI development.

Potential improvements include:

- **Real-time shipment monitoring**
- **Predictive delay analysis**
- **Predictive ETA modeling**
- **Route optimization**
- **Automated data refresh**
- **Automated operational alerts**
- **Machine-learning-based logistics forecasting**
- **Cloud-based data warehouse deployment**
- **Advanced geographic and network analysis**

These improvements can gradually move the solution from descriptive analytics toward predictive and prescriptive analytics.

12. Strategic Outcome

The strategy provides a reusable Business Intelligence foundation for logistics decision-making.

By integrating **Python, PostgreSQL, SQL, and Tableau**, the project transforms fragmented logistics data into structured and actionable information.

The resulting BI solution enables stakeholders to:

- Monitor logistics performance.
- Identify operational bottlenecks.
- Analyze shipment delays.
- Evaluate routes and transportation performance.
- Assess drivers and vehicles.
- Understand customer and financial performance.
- Evaluate weather-related operational risks.
- Make more informed and data-driven decisions.

The solution also provides a strong foundation for future development of **predictive analytics, real-time monitoring, and automated decision support**.